

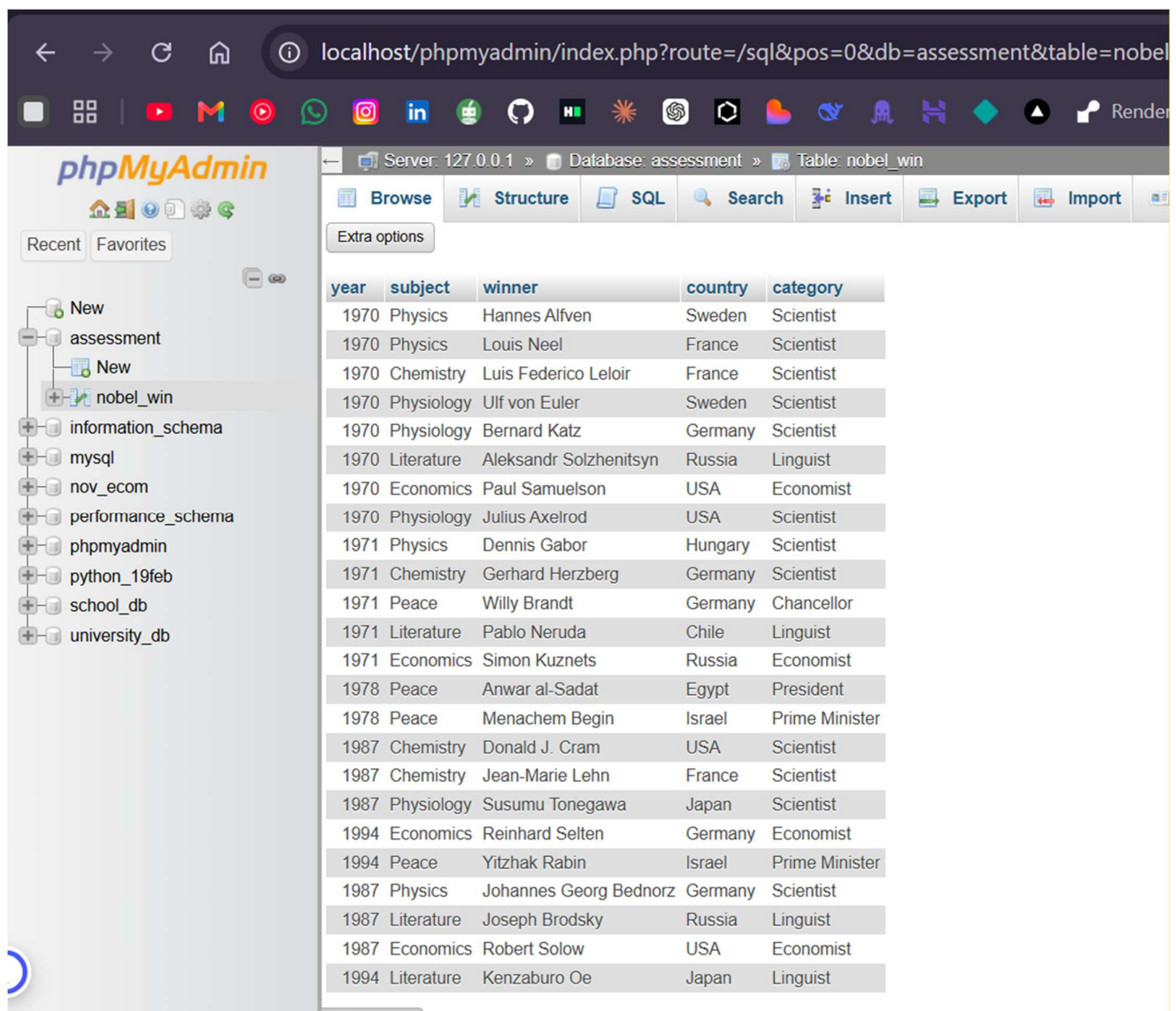
# Assessment-3

## SE - Database Management – SQL & PL/SQL

Q. Write SQL query to solve the problem given below.

Consider a scenario in which you have to create a table called Nobel\_win.

where you have given the information about the entity who has achieved nobel prize in a particular subject all over the world. Over here the table Nobel\_win will contain attributes like year, subject, winner, country and category You need to use the sql commands to create the table given and to solve the problem given below.



The screenshot shows the phpMyAdmin interface for a database named 'assessment'. The table 'nobel\_win' is selected, and its structure and data are displayed. The table has five columns: year, subject, winner, country, and category. The data is as follows:

| year | subject    | winner                 | country | category       |
|------|------------|------------------------|---------|----------------|
| 1970 | Physics    | Hannes Alfvén          | Sweden  | Scientist      |
| 1970 | Physics    | Louis Néel             | France  | Scientist      |
| 1970 | Chemistry  | Luis Federico Leloir   | France  | Scientist      |
| 1970 | Physiology | Ulf von Euler          | Sweden  | Scientist      |
| 1970 | Physiology | Bernard Katz           | Germany | Scientist      |
| 1970 | Literature | Aleksandr Solzhenitsyn | Russia  | Linguist       |
| 1970 | Economics  | Paul Samuelson         | USA     | Economist      |
| 1970 | Physiology | Julius Axelrod         | USA     | Scientist      |
| 1971 | Physics    | Dennis Gabor           | Hungary | Scientist      |
| 1971 | Chemistry  | Gerhard Herzberg       | Germany | Scientist      |
| 1971 | Peace      | Willy Brandt           | Germany | Chancellor     |
| 1971 | Literature | Pablo Neruda           | Chile   | Linguist       |
| 1971 | Economics  | Simon Kuznets          | Russia  | Economist      |
| 1978 | Peace      | Anwar al-Sadat         | Egypt   | President      |
| 1978 | Peace      | Menachem Begin         | Israel  | Prime Minister |
| 1987 | Chemistry  | Donald J. Cram         | USA     | Scientist      |
| 1987 | Chemistry  | Jean-Marie Lehn        | France  | Scientist      |
| 1987 | Physiology | Susumu Tonegawa        | Japan   | Scientist      |
| 1994 | Economics  | Reinhard Selten        | Germany | Economist      |
| 1994 | Peace      | Yitzhak Rabin          | Israel  | Prime Minister |
| 1987 | Physics    | Johannes Georg Bednorz | Germany | Scientist      |
| 1987 | Literature | Joseph Brodsky         | Russia  | Linguist       |
| 1987 | Economics  | Robert Solow           | USA     | Economist      |
| 1994 | Literature | Kenzaburo Oe           | Japan   | Linguist       |

**Q.1. Write sql query to find the nobel prize winners of the year 1970.  
Return year,subject and winner**

**Ans:** SELECT year, subject, winner FROM Nobel\_win WHERE year = 1970;

Showing rows 0 - 7 (8 total, Query took 0.0008 seconds.)

```
SELECT year, subject, winner FROM Nobel_win WHERE year = 1970;
```

☐ Profiling [ [Edit inline](#) ] [ [Edit](#) ] [ [Explain SQL](#) ] [ [Create PHP code](#) ] [ [Refresh](#) ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

| year | subject    | winner                 |
|------|------------|------------------------|
| 1970 | Physics    | Hannes Alfven          |
| 1970 | Physics    | Louis Neel             |
| 1970 | Chemistry  | Luis Federico Leloir   |
| 1970 | Physiology | Ulf von Euler          |
| 1970 | Physiology | Bernard Katz           |
| 1970 | Literature | Aleksandr Solzhenitsyn |
| 1970 | Economics  | Paul Samuelson         |
| 1970 | Physiology | Julius Axelrod         |

☐ Show all | Number of rows: 25 | Filter rows: Search this table

**Q.2. Write sql query to find the nobel prize winners in chemistry between the years 1965 and 1975. Begin and end values are includedReturn year, subject, winner and country**

**Ans:** SELECT year, subject, winner, country FROM Nobel\_win WHERE subject = 'Chemistry' AND year BETWEEN 1965 AND 1975;

Showing rows 0 - 1 (2 total, Query took 0.0007 seconds.)

```
SELECT year, subject, winner, country FROM Nobel_win WHERE subject = 'Chemistry' AND year BETWEEN 1965 AND 1975;
```

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☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

| year | subject   | winner               | country |
|------|-----------|----------------------|---------|
| 1970 | Chemistry | Luis Federico Leloir | France  |
| 1971 | Chemistry | Gerhard Herzberg     | Germany |

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

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**Q.3. Write sql query to retrieve the details of the winners whose first name matches with the string 'Louis'. Return year,subject,winner,country**

**Ans:** SELECT year, subject, winner, country FROM Nobel\_win WHERE winner LIKE 'Louis %';

✓ Showing rows 0 - 0 (1 total, Query took 0.0006 seconds.)

```
SELECT year, subject, winner, country FROM Nobel_win WHERE winner LIKE 'Louis %';
```

☐ Profiling [\[ Edit inline \]](#) [\[ Edit \]](#) [\[ Explain SQL \]](#) [\[ Create PHP code \]](#) [\[ Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

| year | subject | winner     | country |
|------|---------|------------|---------|
| 1970 | Physics | Louis Neel | France  |

☐ Show all | Number of rows: 25 | Filter rows:

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

**Q.4. Write sql query to find Nobel prize winners for the subject that does not begin with the letter 'P'. Order the result by year, descending and winner in ascending**

**Ans:** SELECT \* FROM Nobel\_win WHERE subject NOT LIKE 'P%' ORDER BY year DESC, winner ASC;

✓ Showing rows 0 - 11 (12 total, Query took 0.0006 seconds.) [year: 1994... - 1970...] [winner: KENZABURO OE... - PAUL SAMUELSON...]

```
SELECT * FROM Nobel_win WHERE subject NOT LIKE 'P%' ORDER BY year DESC, winner ASC;
```

☐ Profiling [\[ Edit inline \]](#) [\[ Edit \]](#) [\[ Explain SQL \]](#) [\[ Create PHP code \]](#) [\[ Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

| year | subject    | winner                 | country | category  |
|------|------------|------------------------|---------|-----------|
| 1994 | Literature | Kenzaburo Oe           | Japan   | Linguist  |
| 1994 | Economics  | Reinhard Selten        | Germany | Economist |
| 1987 | Chemistry  | Donald J. Cram         | USA     | Scientist |
| 1987 | Chemistry  | Jean-Marie Lehn        | France  | Scientist |
| 1987 | Literature | Joseph Brodsky         | Russia  | Linguist  |
| 1987 | Economics  | Robert Solow           | USA     | Economist |
| 1971 | Chemistry  | Gerhard Herzberg       | Germany | Scientist |
| 1971 | Literature | Pablo Neruda           | Chile   | Linguist  |
| 1971 | Economics  | Simon Kuznets          | Russia  | Economist |
| 1970 | Literature | Aleksandr Solzhenitsyn | Russia  | Linguist  |
| 1970 | Chemistry  | Luis Federico Leloir   | France  | Scientist |
| 1970 | Economics  | Paul Samuelson         | USA     | Economist |

☐ Show all | Number of rows: 25 | Filter rows:

**Q.5. Write sql query to find the details of 1970 Nobel prize winners. Order the result by subject, ascending except for 'Chemistry' and 'Economics' which will come at the end of the result set. Return year, subject, winner , country and category.**

**Ans:** SELECT year, subject, winner, country, category FROM Nobel\_win WHERE year = 1970 ORDER BY CASE WHEN subject IN ('Chemistry', 'Economics') THEN 2 ELSE 1 END, subject ASC;

✓ Showing rows 0 - 7 (8 total, Query took 0.0008 seconds.) [subject: LITERATURE... - ECONOMICS...]

```
SELECT year, subject, winner, country, category FROM Nobel_win WHERE year = 1970 ORDER BY CASE WHEN subject IN ('Chemistry', 'Economics') THEN 2 ELSE 1 END, subject ASC;
```

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☐ Show all | Number of rows: 25 ▼ Filter rows:

Extra options

| year | subject    | winner                 | country | category  |
|------|------------|------------------------|---------|-----------|
| 1970 | Literature | Aleksandr Solzhenitsyn | Russia  | Linguist  |
| 1970 | Physics    | Hannes Alfvén          | Sweden  | Scientist |
| 1970 | Physics    | Louis Néel             | France  | Scientist |
| 1970 | Physiology | Ulf von Euler          | Sweden  | Scientist |
| 1970 | Physiology | Bernard Katz           | Germany | Scientist |
| 1970 | Physiology | Julius Axelrod         | USA     | Scientist |
| 1970 | Chemistry  | Luis Federico Leloir   | France  | Scientist |
| 1970 | Economics  | Paul Samuelson         | USA     | Economist |

☐ Show all | Number of rows: 25 ▼ Filter rows: